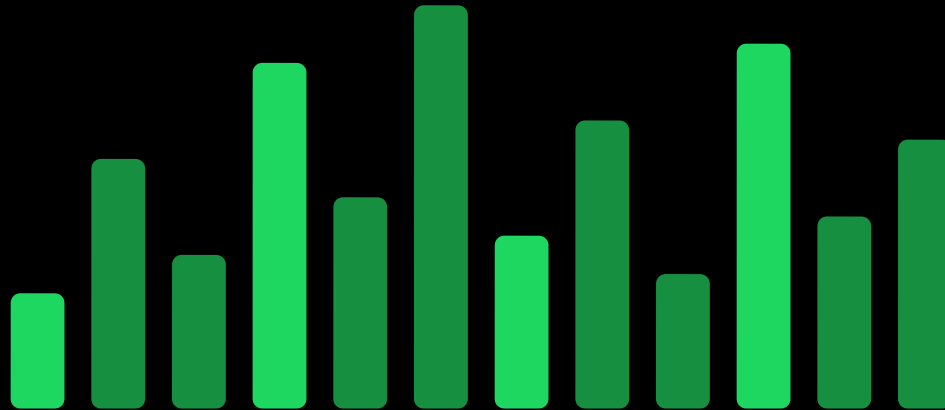


DOES SOUND PREDICT SUCCESS?

A Data-Driven Investigation into What Makes a Spotify Track Chart



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INFS 343 - Business Analytics Final Project



32,828 Real Spotify Tracks

Sourced from Kaggle's 30,000 Spotify Songs dataset, containing audio features, genre, and popularity scores pulled directly from Spotify's API.

32,828

Songs analyzed
(after cleaning)

23

Variables per track

6

Genres
(EDM, Latin, Pop, R&B, Rap, Rock)

5

Rows removed
(missing data)



This data set was mostly clean coming into this project and there was minimal reformatting that needed to be done in terms of data types

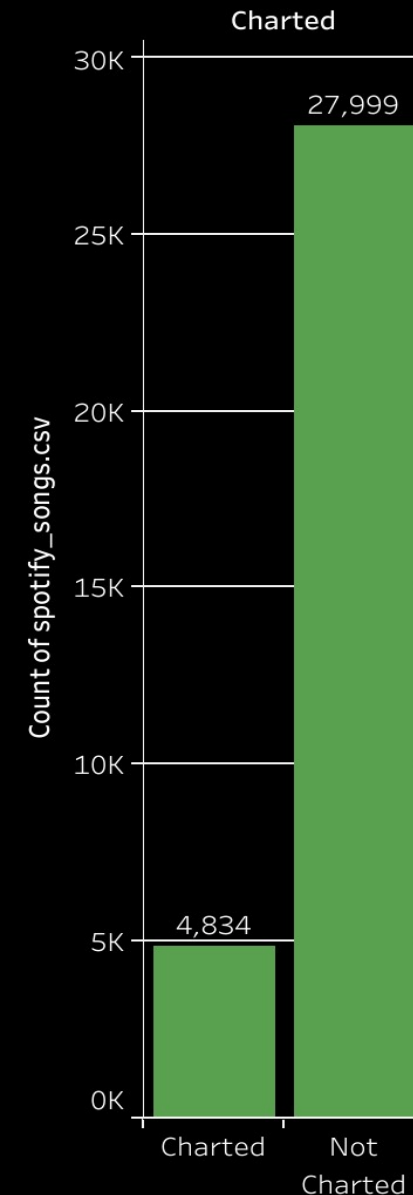
What Counts as “Charting”?

Spotify doesn't track Billboard chart status, so I created my own definition:

charted = **track_popularity $\geq$ 70**

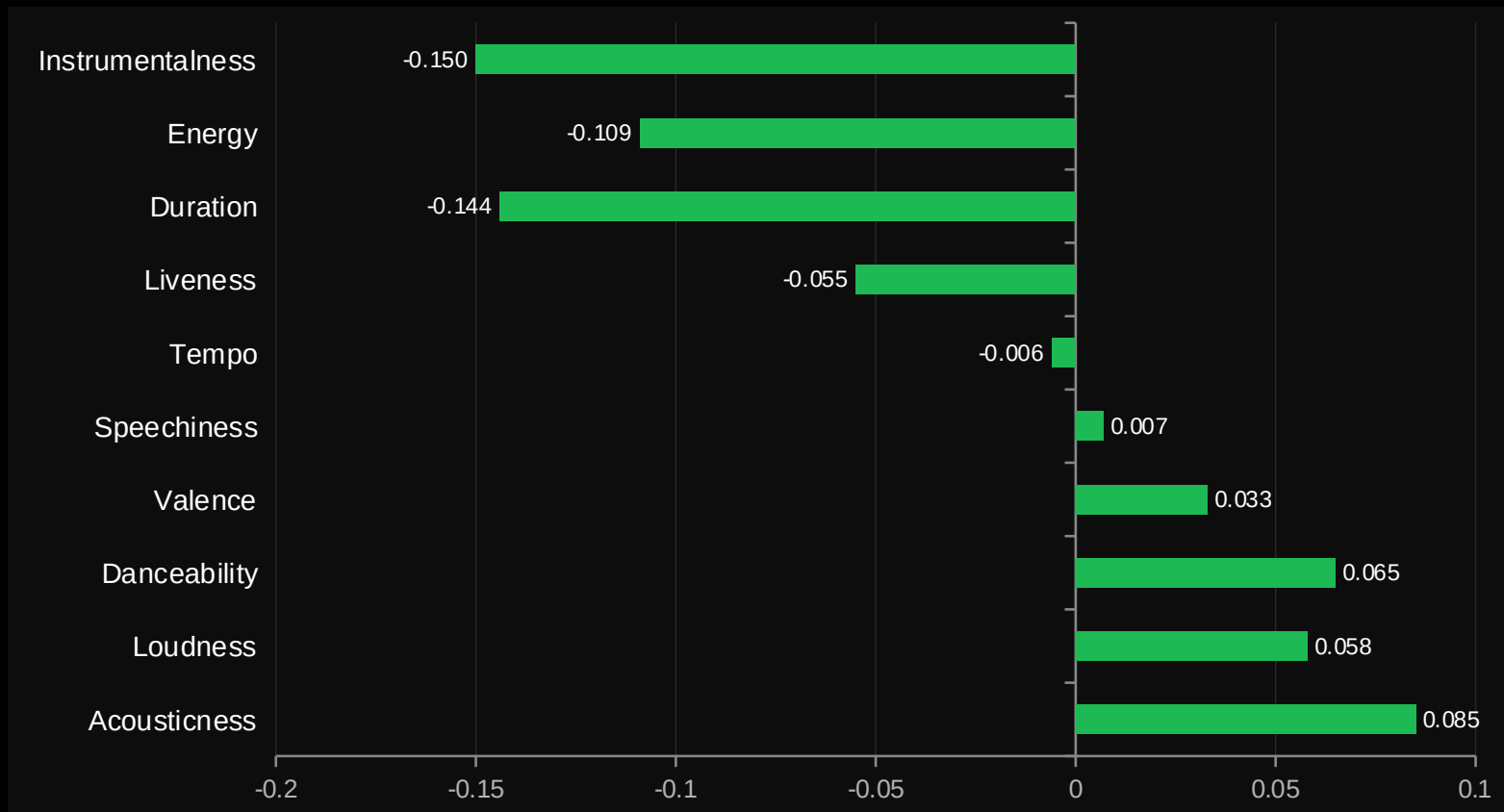


Key challenge spotted early: only ~15% of songs “chart” under this definition, this may lead to issues with the predictive model because the data is heavily imbalanced





Audio Features and their Correlation



STRONGEST 3

Instrumentalness

-0.150

Duration

-0.144

Energy

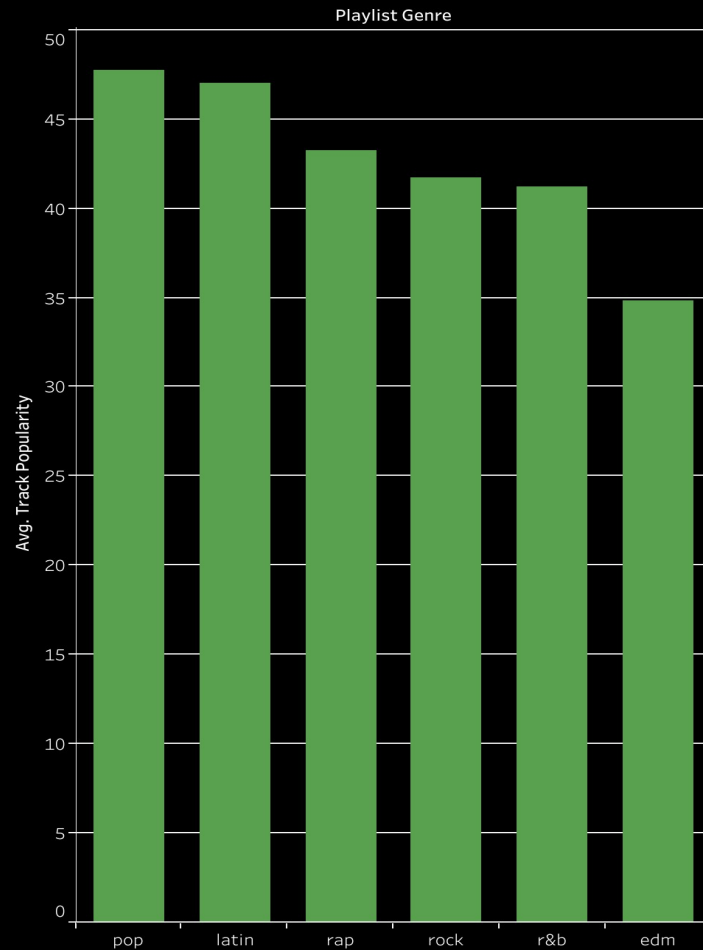
-0.109

Even the "best" predictors barely qualify as weak correlations.

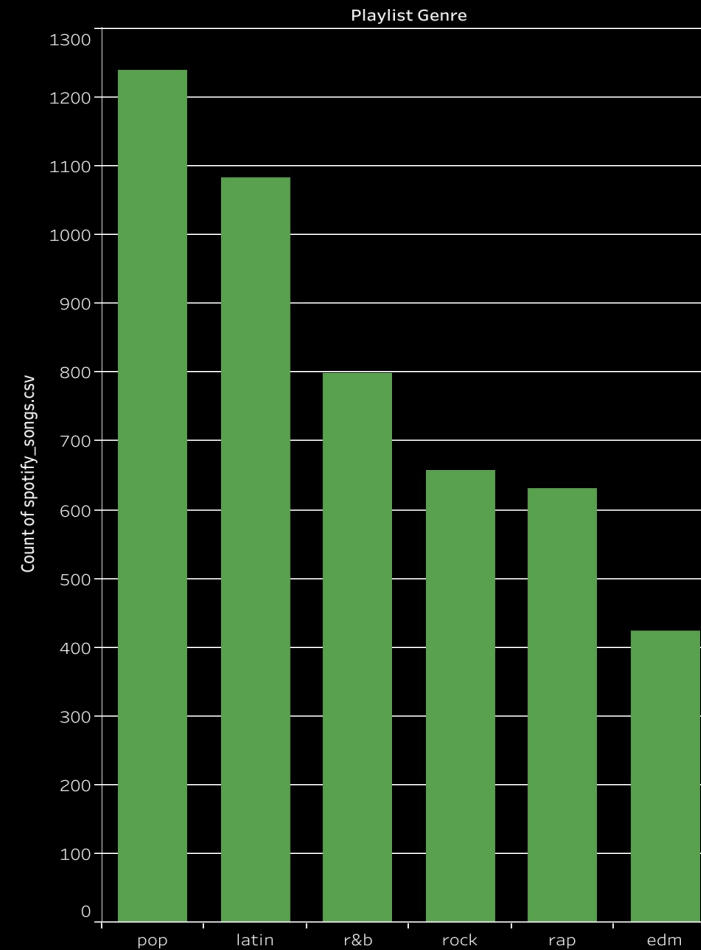


Genre Insights

Genre vs. Popularity



Genre of Charted Songs



Pop & Latin lead

averaging 47+ popularity, vs. EDM's 34.8, a 13-point gap that no single audio feature comes close to explaining.

Genre-charted breakdown:

Pop: 1,240 charted songs (highest)

Latin: 1,083 charted songs

EDM: only 424 charted songs



Decision Tree

Trained on instrumentalness, energy, and duration (70/30 train/test split) to predict charted status.

Root Node Only

No splits found

rpart found no split across instrumentalness, energy, or duration.

85.1%

Reported
accuracy

0

"Yes" predictions
ever made

85.3%

Baseline accuracy
of guessing "No"



Allocating Studio Time for Maximum Impact

Scenario: using illustrative capacity and budget features a label splits limited studio hours and budget between energy-focused (E) and instrumental-focused (I) tracks to maximize total expected popularity.

$$\text{MAX } Z = 40.41(E) + 31.81(I)$$

subject to:

$4E + 6I \leq 160$ (studio hours)

$500E + 300I \leq 12,000$ (budget)

13.33

Energy tracks (E)

17.78

Instrumental tracks (I)

1,104.3

Max popularity score



Dual value insight: one more studio hour adds ~2.10 points of expected popularity, but one more dollar of budget adds only ~0.064 points. **Studio hours are the real bottleneck**, if the label can invest in only one resource, it should be production time, not budget.



Three Takeaways for the Industry

01

Sound alone won't make a hit

Even the strongest audio features (instrumentalness, duration, energy) explains a small percentage of popularity. Charting depends far more on other factors.

02

Genre & context outweigh audio signals

Pop and Latin tracks chart 2-3x more often than EDM, pointing to audience, marketing, and genre as bigger levers than the specific individual features of a song.

03

Optimization still has clear answers

Even without a predictive model, optimization pinpoints the real constraint. Using illustrative budget and capacity figures, studio hours come out ~33x more valuable than extra budget at the margin, this is the kind of bottleneck a label could find by plugging in their own real numbers.

THANK YOU

